

InMobi DSP: An Overview for Engineering Candidates

Intended Audience

This document is aimed at the future employees of iDSP Engineering that are applying for various engineering roles. This document does not assume the candidates to have any background on Advertising Technology Stack or Ad-Tech industry in general.

What you will learn

In this document, you will get an overview of Inmobi Demand Side Platform, aka iDSP or Inmobi DSP, the core business problems iDSP solves for, the interesting engineering problems, the tech stack that powers our iDSP, the team behind iDSP and our typical interview loop.

If you are absolutely noob to ad-tech industry, read on; If you have some familiarity, feel free to skip this section and head directly to “Inmobi Demand Side Platform [iDSP]”.

A Note Before You Begin

This document is the compilation of our pitch to the future engineers of Inmobi as well as the common questions that the candidates have asked us across hundreds of interview conversations. Internalising this information is a must before the interview loop starts. You can expect the interviewer to ask questions **based on** the content of this document.

Advertising

First, the basics.

Every business needs customers — without them, there’s no reason for the business to exist.

A business typically aims to:

- Acquire and retain customers
- Build meaningful engagement
- Promote its products, services, and brand identity

More and more businesses are competing to acquire and engage the same audience. Thus the competition.

Where there’s competition, businesses market their products and services to achieve these objectives — that’s where marketing comes in.

Advertising is a type of Marketing

Let's focus our discussion on digital advertising only. You may have encountered the following forms of digital advertising:

- Social Ads - On Facebook, Instagram, TikTok
- Search Ads - On Google Search Listing
- Display Ads - Banners on Websites and Mobile App
- Audio Ads - On Podcast
- OTT \ Connected TV Ads - Netflix, Disney + HotStar, Hulu
- Video Ads - Ads within Youtube
- Email Ads
- Out of Home Ads - Ones you see in the restaurants or other public places

By the way, you are typically spotted at one of the following digital environments:

- Mobile App [e.g espnCricInfo, Angry Birds, SnapChat]
- Mobile Web Browser [e.g Safari on iOS, Chrome on Android Phones]
- Desktop Web Browser [e.g Chrome on Windows OS]
- Netflix on Sony TV using Android TV \ Google TV
- Listening to Podcasts on Spotify

Programmatic Advertising

Let's switch gears. If you are not living under a rock, you must have seen some type of ads, be it in the form of Video or Banner Image or Text on sites such as Youtube, <http://cnn.com> , <http://ndtv.com> , or on apps such as *Instagram, Candy Crush, Angry Birds, espncricinfo*.

You may have seen ads for quick commerce services [such as **Zepto, Instamart**], eCommerce apps [such as **Myntra**], Food and Beverages [such as **McDonald's, Dominoes**], Games [such as **Monopoly, Clash of Clans**] and so on.

Before we move on, quick FAQs:

Who is paying for these ads?

- Businesses or in our parlance, Advertisers. In the above examples, the ones in bold

Why would apps and websites allow ads?

- Advertising revenue, in simple terms, dollars [\$], is an important revenue stream for most businesses

Is there a term for the apps or websites that show ads?

- Yes, they are typically called as “Supply”, more generally, supply side of Ad market[place]. The italicised names in the above paragraph are the ones that fulfill the supply of ad inventory.

Then, there must be a demand angle to this?

- Yes, the demand part of the equation is fulfilled by advertisers that buy impressions. An Impression is an opportunity to show their ads to the end user. In short, advertisers are the demand people.

Are we talking about some sort of marketplace here?

- Yes, Digital Programmatic Advertising is a marketplace - where the suppliers provide the space for ad-inventory, the demand i.e the advertisers buy the ad impression and a middle entity called as Exchange that connects the supply and demand enabling the purchase of ad impressions in real time.

Keep it simple, Tell me more about this Exchange?

- Sure. Think of a **stock exchange** (like the New York Stock Exchange), but instead of buying and selling shares of companies, people are buying and selling **ad space** on websites and apps. That's an ad exchange in a nutshell.

 An **Ad Exchange** is the high-speed, digital marketplace where **ad space is automatically auctioned off to the highest bidder in the blink of an eye.**

Help me with an example please?

Sure, let me explain the mechanism in 3 simple steps

How it Works in 3 Simple Steps

Let's follow Vijay, who is visiting his favorite cricket news website.

1. **An Ad Space is Available:** As Vijay clicks on a news article, the website has a blank ad banner ready to be filled. It instantly sends a message to the ad exchange: *"I have one ad impression for a user in Mumbai who is a huge cricket fan. Who wants to show him an ad?"*

2. **The Instant Auction:** The ad exchange immediately alerts a crowd of automated advertisers (using their DSPs). Several relevant brands—like a sportswear company, a streaming service showing live matches, and a fantasy cricket app—instantly analyze the opportunity and place their bids.
3. **The Ad is Delivered:** The ad exchange picks the highest bidder in milliseconds. The winner's ad (for example, a flashy new cricket bat) instantly loads onto the webpage, right as the article finishes loading for Vijay.

Programmatic Advertising, in a nutshell, is

- **Automated:** No humans making phone calls or signing contracts for a single ad spot. It's software talking to other software.
- **Hyper-Targeted:** Ads are shown to specific *people* based on their interests and behavior, not just on specific *websites*.
- **Efficient:** Advertisers don't waste money showing ads to people who aren't interested, and publishers (website owners) can sell their ad space for the best possible price.
- **Fast:** The auctions happen in real-time (this is called **Real-Time Bidding** or RTB).

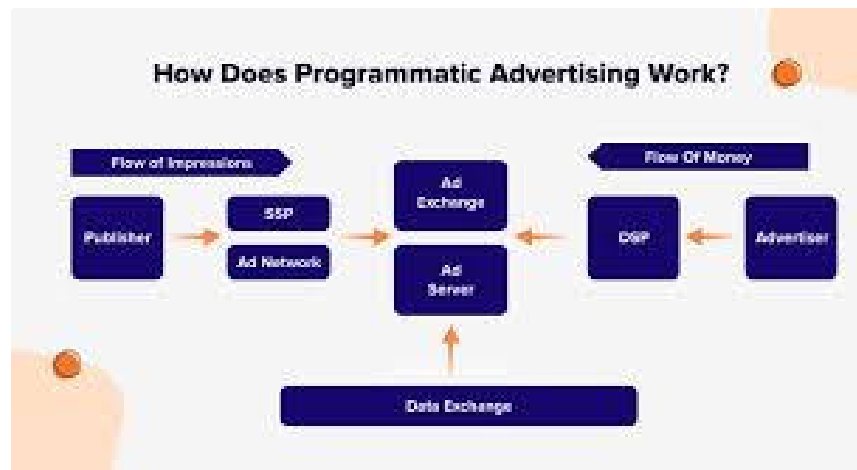
What are the systems involved in the programmatic advertising?

To complete the picture, it helps to know who is talking to the exchange:

- **Advertiser:** The one who wants to show an ad (e.g., Nike).
- **On the Buyer's Side: DSP (Demand-Side Platform)**
 - This is the software *advertisers* use. It's their "bidding robot." The advertiser tells their DSP who to target, and the DSP connects to multiple ad exchanges to automatically bid on those people.
- **On the Seller's Side: SSP (Supply-Side Platform)**
 - This is the software *website owners* use. It's their "sales robot." The SSP takes all their ad space and connects it to multiple ad exchanges to get the highest possible price from advertisers.

Uber level architecture view of Programmatic Advertising system

Courtesy: <https://medium.com/@akankshachandan/what-is-programmatic-advertising-22d78a793fc5>



Inmobi Demand Side Platform [iDSP]

Let's focus our discussion on Demand-Side Platform, specifically iDSP.

Who are the customers of iDSP? What do they want? What do they pay, and measure the performance of iDSP?

iDSP's customers are advertisers. Any business that want or have customers can be an advertiser.

Advertisers typically want to promote their products and services, to acquire new customers, retain their customers by re-engaging\re-marketing to their current customers.

Advertisers pay dollars [money] in the form of ad-budget. This is the spend or the investment from advertisers.

There are many metrics to measure the Return-On-Investment [ROI], depending on the objective of the advertisers such as Return-on-ad-spend [ROAS], Cost-per-acquisition [CPA], Cost-per-install [CPI], Leads generated, App Installs, Website Traffic, Newsletter sign-ups etc.

Engineering @ Inmobi Demand Side Platform [iDSP]

How does iDSP achieve the above mentioned business outcomes?

iDSP is composed of few critical complex decisioning systems that are low latent [<100 msec] and operate at the largest of scale [X million requests per second throughout the day] targeting the right users at the right time to meet the advertiser's objectives.

iDSP, in a gist, consists of

- **Core Real-Time-Bidder engine** that computes the bid value; More broadly, values the qualified incoming request per advertiser objectives.
- **Targeting Engine** - to find, filter and score the relevant users per advertiser's campaigns
- **Creatives\Ad-Experience Engine** - to select the right ad experience to show to the user
- **Supply Quality Engine** - that filters out the incoming requests basis the quality of supply
- **Fraud Control Engine** - that filters the fraudulent or suspicious incoming ad requests

Note that most of the above decisioning engines are highly Machine-Learning [ML] driven. ML works well at scale.

By the way, what's the scale?

iDSP processes few million requests per second, not at the lower end of a million though, throughout the day. There is no defined peak unlike other engineering systems. Exchanges expect the demand participants, i.e DSPs, to return with an ad-response [or none] in less than 100-200 milliseconds.

We run one of the largest data and ML stack, supporting few million ML inferencing per second, TBs of data per hour, hundreds of data engineering and ML training jobs are being orchestrated every day.

What we're building @ iDSP?

Since 18 months, we've been on a mission transforming our iDSP to ML-driven iDSP platform. Almost all the critical decisioning at high scale are being implemented by one or more machine learning models.

We absolutely believe that adopting AI, ML and Data Science techniques to solve DSP business problems will yield the sustainable and scalable results over time.

What are the engineering problems that are solved @ iDSP?

Diverse and wide varieties of engineering problems to support our mission. A short list follows:

- ML Experimentation consisting of A/B framework, Model Monitoring, Drift Detection, Feature Store, Offline Training and Online Inferencing at scale
- Fin-Ops - Operating our huge infra within cost envelope

- Ad-serving at planet scale
- Orchestrating hundreds of data engineering jobs crunching TBs of data per hour
- Privacy, Governance and Compliance
- Algorithms to select right Ad-experiences
- Optimization of data engineering and ML training
- Plethora of system and business health dashboards
- Engineering and Operational Excellence at our scale

What is our tech stack?

- All our serving stack is predominantly follow micro services architecture. Java, Scala, SpringBoot are our go to languages and frameworks here.
- We love Aerospike.
- We do use SQL for the use cases that demand strong consistency.
- Apache Flink for our streaming use cases.
- In the data world, it's Spark every where. Python and Scala will take you to places. AirFlow is our orchestrator. We use a third party notebook solution for Exploratory Data Analysis [EDA].
- For offline training, we use MLFlow. We use variety of frameworks such as PyTorch, TensorFlow, Keras.
- For online inferencing, we use open source frameworks such as TF-serving for Neural Net Models and engines such as Triton for inferencing.
- For OLAP and BI use cases, we use Spark, Trino, ClickHouse.

We have totally embraced and stretching the contemporary Vibe coding tools such as Cursor, Claude Code. Be ready to showcase your skills there and teach us a technique or two during our conversations.

What is the team like?

- About 30+ engineering members mostly rolling up to two engineering leaders. Barring a couple that work on Advertiser UI, most of the engineers work across the serving, data and ML stack, across the technology frameworks and languages.
 - About 10+ Data Scientists, few PhDs, few master degree holders and bunch of seasoned scientists in advertising landscape.
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Explain the role of SDE IV \ Technical Lead and above [Staff Engineers..]

- You are an individual contributor
- You must have demonstrated history of operating with high agency and ownership
- Learnability is key to thrive at iDSP
- A great problem solver; You are product builder.
- Should work across the technology stack solving for business problems
- Demonstrated high bar of engineering and operational excellence
- You are fundamentally a good compute science engineer; Every thing else can be built on top of the fundamentals.
- You are a Talent Builder; You will play a key role in building the pipeline of great engineers; You will be a mentor, and a coach for many budding engineers.
- You will write well, exceptionally well — with clarity, coherence, and conciseness
- You will partner well. Your partners are Data Scientists, Engineering Leaders, Product Managers, Product Analysts, and Operations Team [Traders].

What's our interview loop like ?

Typically, following rounds

- Machine Coding
- Data Structure and Algorithms
- Low Level Design
- System Design
- Problem Solving
- Bar Raiser
- Hiring Manager

For Staff and Senior members, you will meet our HR Business Partner team too.

What should I carry to the interview?

- Curiosity
- Questions. Ask questions to the interviewer to know more about Inmobi Ads. Do the groundwork based on the above information.
- Share information about what you have learnt about Ad-tech ecosystem. Gain Brownie points.

 Our Interviews are Conversations with the potential future colleague. Not a mere Q and A session. We learn from you and your methods too.

Wear your problem solver hat, **Good Luck !**